IMPERIAL COLLEGE LONDON

EE4-48
EE9-FPN1-03

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE PART IV: MEng and ACGI

POWER SYSTEM DYNAMICS, STABILITY AND CONTROL

Thursday, 14 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : B. Chaudhuri

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The Questions

1.

a)

i) What are the defining parameters in modelling a lightning surge in power networks?

[2]

ii) What components design does the lightning surge influence?

[2]

iii) Why are usually the stator windings of a synchronous generator star connected?

[2]

b)

i) Briefly narrate the nature and importance of transient or first swing stability in the context of power system operation.

[4]

ii) What are the modelling considerations for analysing transient stability problem in power systems?

[4]

iii) How are these modelling considerations influenced by the increasing presence of renewable generation in the grid?

[4]

c)

i) What is the purpose of speed governing in a grid-connected generator?

[3]

ii) Why is it relatively difficult to design the speed governing system for a hydro turbine coupled with a large synchronous generator?

[4]

[25 marks]

2.

a)

i) Why is the rotor of a synchronous generator made of solid construction whereas its stator is laminated?

[3]

ii) What is the purpose of closed-circuit amortisseur windings in synchronous machine rotor?

[3]

iii) Why such windings are not present in wind turbine rotor?

[3]

iv) How are amortisseur windings modelled in power system studies?

[3]

v) A round rotor synchronous generator has a synchronous reactance of 2.5 pu on its own base. When a sudden short circuit appears at the terminal with rated pre-fault voltage – the initial current is 4-5 times the rated current. It lasts for about 5 seconds before falling gradually. Why is the generator initial output current so high?

[4]

vi) A synchronous generator is operating at full load with rated excitation. For some reason excitation system monitoring sends an alert; the level of excitation is reduced by 10%. How will it affect the transient stability of the generator?

[3]

vii) What is the purpose of d-q transformation in synchronous machine modelling and analysis?

[3]

viii) Two synchronous generators of identical capacity are delivering full load to the grid. One of them has uniform air gap between stator and rotor and the other does not. For some reason, the loss of excitation relay operates in both. Which generator must have faster and stronger control to prevent going out of synchronism and why?

[1+2]

[25 marks]

3.

a)

Fig Q3.1 shows a simple power system model with one synchronous generator connected to a power network through a generator transformer and a section of power transmission line to deliver real and reactive power to the system. The values of various operating variables and parameters shown in the diagram are as follows.

E = 3.2 pu, Vt = 1.0 p.u, Xs= 2.7 p.u, Xt =0.08 p.u, Xl= 0.15 p.u , δ = 30 degree.

i) Ignoring stator resistance, derive the power angle equations for both active and reactive power from the first principle.

[4]

ii) With the help of the derived power angle equations obtained in 3 a) (i) compute the real and reactive power at the network end.

[2]

b)

Because of a transient disturbance in the system, the network voltage dropped slightly and triggering the electromechanical dynamics of the generator connected to the system. Assuming a classical swing equation model (speed $\omega_r$and load angle δ as only dynamic variables) with an inertia constant H (sec) and damping constant D (in pu);

i) Write down the swing equations.

[2]

ii) For the values of the operational variables listed in 3(a) and constants listed below, obtain the linearized dynamics in state-space form treating mechanical power as input and generator real power as output.

H = 5.0 sec, D = 10.0 pu, operating frequency is 50 Hz or 314 rad/sec.

[4]

(iii) With the help of the state, input and output matrix obtain the transfer function polynomial in 's' with all numerical coefficients.

[8]

iii) Comment on the small-signal stability of the system with the help of the roots of the denominator polynomial obtained in 3 b) (iii).

[5]

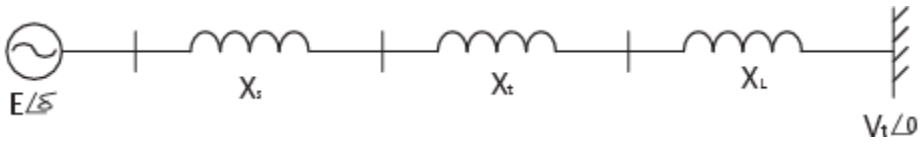


Fig Q3.1 A Simple Power System

[25 marks]

4.

a)

i) List two popular wind power generation technologies out of all possible ones that have been tried in the wind power industry.

[2]

ii) From the perspective of stability of operation, what will be the features for a design and control engineer to focus on a popular technology of generation?

[4]

iii) Which of the transient and small-signal stability will be more important for a typically sized wind turbine generator and why?

[1+3]

b)

ii) With the help of power output v/s wind speed define, cut-in, rated and cut-out wind speed of a wind turbine generator (WTG)

[3]

ii) Mention traditional methods of WTG control in all speed range of operation.

[3]

iii) What is maximum power point tracking in relation to the operation of a WTG?

[2]

c)

i) What is the performance coefficient (C_p) of a wind turbine?

[2]

ii) What are the factors influencing the performance coefficient (C_p)?

[2]

iii) What maximum percentage of power from the wind can be converted to mechanical form?

[1]

(iv) What is the significance of*wake* in the context of wind farm operation?

[2]

[25 marks]